

SAGRES, Portugal

WMO: 085330

Lat: 37.0128N

Lon: 8.9489W

Elev: 85

StdP: 14.65

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.1	42.7	27.8	21.8	46.1	32.8	27.3	47.2	27.7	54.6	24.7	55.0	4.4	70	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.8	84.0	67.4	80.9	67.1	78.2	66.5	70.6	77.5	69.3	76.1	68.4	74.7	8.2	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.4	104.8	72.9	67.3	100.6	71.8	66.3	97.1	71.1	34.5	77.6	33.4	76.4	32.7	74.9	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.0	22.1	20.3	DB	35.8	90.8	3.0	4.9	33.6	94.3	31.8	97.2	30.1	99.9	27.9	103.5	(4)
(5)			WB	32.9	72.6	3.1	1.6	30.6	73.8	28.8	74.7	27.0	75.6	24.7	76.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.3	54.0	54.6	57.7	59.9	63.7	67.7	69.2	70.3	68.9	65.5	59.4	55.9
	DBStd	6.83	4.19	3.83	3.44	3.15	3.69	3.58	3.19	3.40	2.87	3.55	4.32	4.26
	HDD50	28	13	8	2	0	0	0	0	0	0	0	1	4
	HDD65	1592	340	290	227	157	69	11	1	0	3	36	175	283
	CDD50	4507	137	138	240	296	425	530	596	630	567	480	281	187
	CDD65	597	0	0	0	2	28	91	132	165	121	52	6	0
	CDH74	1926	0	0	2	11	92	297	454	651	335	84	0	0
	CDH80	371	0	0	0	0	14	53	103	156	39	6	0	0
Wind	WSAvg	9.6	8.2	9.0	8.9	9.7	10.3	11.2	13.2	11.9	8.5	7.6	8.0	8.5
Precipitation	PrecAvg	18.1	3.1	2.7	1.7	1.5	0.7	0.3	0.0	0.2	0.5	2.2	2.6	2.6
	PrecMax	27.9	6.7	8.2	4.1	3.8	2.6	2.0	0.0	1.9	2.8	8.4	8.4	8.2
	PrecMin	10.0	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0
	PrecStd	4.8	2.0	1.9	1.1	1.2	0.7	0.5	0.0	0.4	0.8	2.4	2.1	2.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.1	65.1	70.9	75.4	80.9	85.8	88.2	88.5	84.1	80.1	71.4	65.9
		MCWB	59.3	58.4	59.0	61.9	63.3	66.3	67.1	67.7	66.1	65.1	64.6	61.3
	2%	DB	63.1	63.3	67.3	70.8	76.0	80.4	82.6	83.9	80.5	76.0	69.5	64.8
		MCWB	58.7	57.9	58.4	60.0	63.0	65.9	67.6	68.1	67.3	65.8	63.9	61.3
	5%	DB	62.0	62.0	65.3	68.1	73.0	77.0	78.9	80.7	77.8	73.5	68.1	63.9
		MCWB	57.8	57.2	58.3	59.9	62.2	65.6	66.8	68.4	66.7	66.1	63.0	60.7
	10%	DB	60.9	61.0	63.7	66.1	70.2	74.3	75.5	77.1	75.2	71.5	66.6	62.9
		MCWB	57.0	56.6	57.9	59.5	62.1	65.2	66.2	67.3	66.5	65.2	61.8	59.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.3	61.0	62.3	64.6	67.5	70.1	70.9	72.8	71.7	70.3	67.8	63.9
		MCDB	63.8	62.9	65.8	70.4	74.0	76.6	80.5	80.4	78.0	73.5	69.9	64.7
	2%	WB	60.5	59.7	61.1	62.6	65.5	68.3	69.1	70.7	70.1	68.7	66.1	62.8
		MCDB	61.9	61.9	64.2	67.7	71.5	75.8	78.1	78.6	75.7	72.2	68.5	64.2
	5%	WB	59.2	58.6	60.1	61.2	64.2	67.1	67.9	69.3	68.9	67.5	64.3	61.6
		MCDB	61.1	61.1	63.3	66.1	69.8	74.3	75.5	77.1	74.3	71.2	66.8	63.4
	10%	WB	57.9	57.5	59.1	60.2	63.0	65.9	66.8	68.1	67.8	66.4	62.8	60.1
		MCDB	60.5	60.5	62.5	64.9	68.3	72.5	73.7	75.2	73.1	70.2	65.6	62.4
Mean Daily Temperature Range	5% DB	MDBR	12.0	11.6	11.3	11.0	11.4	12.0	12.3	12.8	12.4	11.4	11.8	11.4
		MCDBR	11.9	11.9	13.7	14.4	16.2	16.8	17.8	18.3	16.8	14.1	11.5	10.1
		MCWBR	9.0	8.5	8.6	8.6	8.2	7.1	7.7	8.5	8.4	8.2	8.1	7.9
	5% WB	MCDBR	7.9	9.5	10.7	12.1	12.3	13.6	14.2	14.9	13.2	11.0	8.5	7.1
		MCWBR	7.2	7.6	7.7	8.0	7.0	6.8	6.8	7.8	7.8	7.4	7.0	6.6
Clear-Sky Solar Irradiance	taub	0.336	0.351	0.385	0.379	0.386	0.390	0.377	0.396	0.389	0.375	0.349	0.340	
	taud	2.514	2.475	2.380	2.421	2.426	2.430	2.453	2.419	2.454	2.492	2.523	2.505	
	Ebn at Noon	270	280	279	286	285	282	285	278	274	268	263	260	
	Edh at Noon	25	30	36	36	36	36	35	36	33	29	25	24	
All-Sky Solar Radiation	RadAvg	812	1122	1515	1903	2239	2399	2432	2192	1763	1255	890	729	
	RadStd	87	121	124	107	119	126	83	72	92	88	86	72	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page